In [1]:
```
import pandas as pd
import numpy as np
```

In [2]:
```
df = pd.read_csv("claimants.csv")
df.head(10)
```

Out[2]:

	CASENUM	ATTORNEY	CLMSEX	CLMINSUR	SEATBELT	CLMAGE	LOSS
0	5	0	0.0	1.0	0.0	50.0	34.940
1	3	1	1.0	0.0	0.0	18.0	0.891
2	66	1	0.0	1.0	0.0	5.0	0.330
3	70	0	0.0	1.0	1.0	31.0	0.037
4	96	1	0.0	1.0	0.0	30.0	0.038
5	97	0	1.0	1.0	0.0	35.0	0.309
6	10	0	0.0	1.0	0.0	9.0	3.538
7	36	0	1.0	1.0	0.0	34.0	4.881
8	51	1	1.0	1.0	0.0	60.0	0.874
9	55	1	0.0	1.0	0.0	NaN	0.350

Initial Data Analysis

In [3]:
```
df.shape
```

Out[3]:
```
(1340, 7)
```

In [4]:
```
df.dtypes
```

Out[4]:
```
CASENUM       int64
ATTORNEY      int64
CLMSEX      float64
CLMINSUR    float64
SEATBELT    float64
CLMAGE      float64
LOSS        float64
dtype: object
```

In [5]:
```
df.isnull().sum()
```

Out[5]:
```
CASENUM       0
ATTORNEY      0
CLMSEX       12
CLMINSUR     41
SEATBELT     48
CLMAGE      189
LOSS          0
dtype: int64
```

In [6]:
```
df.duplicated().sum()
```

Out[6]:
```
np.int64(0)
```

In [7]:
```
df.describe()
```

Out[7]:

	CASENUM	ATTORNEY	CLMSEX	CLMINSUR	SEATBELT	CLMAGE	LOSS
count	1340.000000	1340.000000	1328.000000	1299.000000	1292.000000	1151.000000	1340.000000
mean	11202.001493	0.488806	0.558735	0.907621	0.017028	28.414422	3.806307
std	9512.750796	0.500061	0.496725	0.289671	0.129425	20.304451	10.636903
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	4177.000000	0.000000	0.000000	1.000000	0.000000	9.000000	0.400000
50%	8756.500000	0.000000	1.000000	1.000000	0.000000	30.000000	1.069500
75%	15702.500000	1.000000	1.000000	1.000000	0.000000	43.000000	3.781500
max	34153.000000	1.000000	1.000000	1.000000	1.000000	95.000000	173.604000

In [8]:
```
df.nunique()
```

Out[8]:
```
CASENUM     1283
ATTORNEY       2
CLMSEX         2
CLMINSUR       2
SEATBELT       2
CLMAGE        68
LOSS         916
dtype: int64
```

In [9]:
```
cat_cols = ["CLMSEX", "CLMINSUR", "SEATBELT"]
num_cols = ["CLMAGE", "LOSS"]
target = "ATTORNEY"
print("Numerical:", num_cols)
print("Categorical:", cat_cols)
print("Target:", target)
```

```
Numerical: ['CLMAGE', 'LOSS']
Categorical: ['CLMSEX', 'CLMINSUR', 'SEATBELT']
Target: ATTORNEY
```

In [10]:
```
df[target].value_counts()
```

Out[10]:
```
ATTORNEY
0    685
1    655
Name: count, dtype: int64
```

Data Pre-processing

In [11]:
```
for c in cat_cols:
    df[c] = df[c].fillna(df[c].mode()[0])
for c in num_cols:
    df[c] = df[c].fillna(df[c].median())
df = df.drop_duplicates()
df.isnull().sum()
```

Out[11]:
```
CASENUM     0
ATTORNEY    0
CLMSEX      0
CLMINSUR    0
SEATBELT    0
CLMAGE      0
LOSS        0
dtype: int64
```

In [12]:
```
df = df.drop(columns="CASENUM")
df.shape
```

Out[12]:
```
(1340, 6)
```

In [13]:
```
Q1 = df[num_cols].quantile(0.25)
Q3 = df[num_cols].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
```

In [14]:
```
outliers = df[((df[num_cols] < lower_bound) | (df[num_cols] > upper_bound)).any(axis=1)]
outliers.shape
```

Out[14]:
```
(69, 6)
```

In [15]:
```
df[num_cols] = df[num_cols].clip(lower_bound, upper_bound, axis=1)
df[num_cols].describe()
```

Out[15]:

	CLMAGE	LOSS
count	1340.000000	1340.000000
mean	28.627612	2.369601
std	18.791557	2.570479
min	0.000000	0.000000
25%	10.000000	0.400000
50%	30.000000	1.069500
75%	40.000000	3.781500
max	85.000000	8.853750

In [16]:
```
df[cat_cols] = df[cat_cols].astype(int)
df.head()
```

Out[16]:

	ATTORNEY	CLMSEX	CLMINSUR	SEATBELT	CLMAGE	LOSS
0	0	0	1	0	50.0	8.85375
1	1	1	0	0	18.0	0.89100
2	1	0	1	0	5.0	0.33000
3	0	0	1	1	31.0	0.03700
4	1	0	1	0	30.0	0.03800

In [17]:
```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
df[num_cols] = scaler.fit_transform(df[num_cols])
df.head()
```

Out[17]:

	ATTORNEY	CLMSEX	CLMINSUR	SEATBELT	CLMAGE	LOSS
0	0	0	1	0	1.137765	2.523487
1	1	1	0	0	-0.565764	-0.575439
2	1	0	1	0	-1.257822	-0.793767
3	0	0	1	1	0.126295	-0.907796
4	1	0	1	0	0.073059	-0.907407

Feature Selection

In [18]:
```
corr = df.corr()[target].drop(target).sort_values()
corr
```

Out[18]:
```
LOSS       -0.408248
SEATBELT   -0.055848
CLMAGE      0.012471
CLMSEX      0.076564
CLMINSUR    0.076631
Name: ATTORNEY, dtype: float64
```

In [19]:
```
features = corr[corr.abs() > 0.05].index.tolist()
features
```

Out[19]:
```
['LOSS', 'SEATBELT', 'CLMSEX', 'CLMINSUR']
```

Logistic Regression

In [20]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt
```

In [21]:
```
X = df[features]
y = df[target]
```

In [22]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
print(X_train.shape, X_test.shape)
```

```
(1072, 4) (268, 4)
```

In [23]:
```
model = LogisticRegression()
model.fit(X_train, y_train)
```

Out[23]:

▾ LogisticRegression

▸ Parameters

▸ Fitted attributes

In [24]:
```
pd.DataFrame({"Feature": features, "Coefficient": model.coef_[0]})
```

Out[24]:

	Feature	Coefficient
0	LOSS	-1.073594
1	SEATBELT	-0.749637
2	CLMSEX	0.362621
3	CLMINSUR	0.359115

In [25]:
```
model.intercept_
```

Out[25]:
```
array([-0.69623239])
```

Prediction

In [26]:
```
y_pred = model.predict(X_test)
```

Evaluation

In [27]:
```
accuracy_score(y_test, y_pred)
```

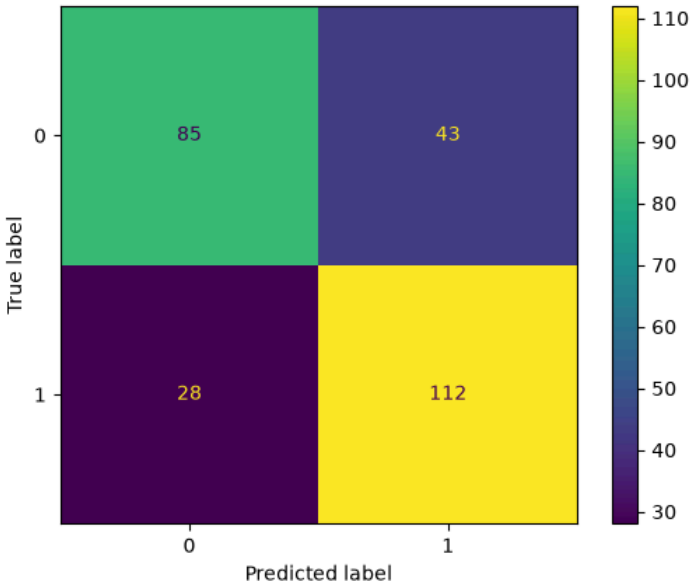
Out[27]:
```
0.7350746268656716
```

In [28]:
```
cm = confusion_matrix(y_test, y_pred)
pd.DataFrame(cm, index=["Actual 0", "Actual 1"], columns=["Predicted 0", "Predicted 1"])
```

Out[28]:

	Predicted 0	Predicted 1
Actual 0	85	43
Actual 1	28	112

In [29]:
```
ConfusionMatrixDisplay(cm).plot()
plt.show()
```



In [30]:
```
comparison = pd.DataFrame({"Actual": y_test.values, "Predicted": y_pred})
comparison.head(15)
```

Out[30]:

	Actual	Predicted
0	0	0
1	0	0
2	1	1
3	0	1
4	0	0
5	0	0
6	1	1
7	0	1
8	0	1
9	0	0
10	1	1
11	0	1
12	1	1
13	0	1
14	1	1